



Cloud Computing

BSE-VB

Submitted By

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Submitted to

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LAB-07

Lab 7 – Environment Variables, PATH, UFW, and SSH Key Authentication

Task 1 – Print and Filter Environment Variables

Steps:

1. Open terminal and display all environment variables.

Screenshot name: task1_printenv_all.png

```
tooba65@ubuntu:~$ printenv
SHELL=/bin/bash
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29t2SagMDAwMDAgMjAwMDAwMAA=
XDG_SEAT=seat0
PWD=/home/tooba65
LOGNAME=tooba65
XDG_SESSION_TYPE=tty
SYSTEMD_EXEC_PID=1197
HOME=/home/tooba65
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:01;cd=40;33:01;or=40;31:01;mi=00:su=37;41:sg=30;43:ca=00:tx=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txi=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01:*.r=01;31:*.cab=01;31:*.wim=01;31:*.sum=01;31:*.dum=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.x01;35:*.cgm=01;35:*.cmf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.m=00;90:*.bak=00;90:*.crdownload=00;90:*.dkg=00;90:*.dkg-old=00;90:*.dkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=1ea15bf1537442a9a5749bca1de84922
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=tooba65
SHLVL=1
XDG_VTNR=1
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/ghome:/usr/local/share:/usr/share:/var/lib/napd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

2. Filter and display only SHELL, HOME, and USER variables.

Screenshot name: task1_grep_shell_home_user.png

```
tooba65@ubuntu:~$ printenv | grep -E "SHELL|HOME|USER"
SHELL=/bin/bash
HOME=/home/tooba65
USER=tooba65
tooba65@ubuntu:~$
```

Task 2 – Create Temporary DB_* Environment Variables

Steps:

1. Create three temporary environment variables for DB_URL, DB_USER, and DB_PASSWORD.

Screenshot name: task2_exports_all.png

```
tooba65@ubuntu:~$ export DB_URL="postgres://db.example.local:5432/mydb"
tooba65@ubuntu:~$ export DB_USER="labuser"
tooba65@ubuntu:~$ export DB_PASSWORD="labpass123"
tooba65@ubuntu:~$
```

2. Display each of the three variables using echo.

Screenshot name: task2_echoes_all.png

```
tooba65@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$ echo "$DB_USER"
labuser
tooba65@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
tooba65@ubuntu:~$
```

3. Verify they are visible in environment variables using filter.

Screenshot name: task2_printenv_grep_db.png

```
tooba65@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
tooba65@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$
```

4. Restart terminal and confirm the variables are no longer available.

Screenshot name: task2_after_restart_checks.png

```
tooba65@ubuntu:~$ echo "$DB_URL"
tooba65@ubuntu:~$
tooba65@ubuntu:~$ printenv | grep '^DB_'
tooba65@ubuntu:~$
```

Task 3 – Make Variables Persistent Using ~/.bashrc

Steps:

1. Open .bashrc file in editor and add the DB variables inside it.

Screenshot name: task3_bashrc_added.png

```
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

# Lab 7 persistent DB variables
export DB_URL="postgres://db.example.local:5432/mydb"
export DB_USER="labuser"
export DB_PASSWORD="labpass123"

-- INSERT --
```

2. Reload the .bashrc file and display the variables again.

Screenshot name: task3_source_and_verification.png

```
tooba65@ubuntu:~$ source ~/.bashrc
tooba65@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$ echo "$DB_USER"
labuser
tooba65@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
tooba65@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$
```

3. Restart terminal again to verify the variables persist.

Screenshot name: task3_after_restart_persistent.png

```
Swap usage: 0%

=> / is using 93.4% of 9.75GB

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
4 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

tooba65@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
tooba65@ubuntu:~$
```

Task 4 – System-Wide Variable, PATH Update, and Welcome Script

Steps:

1. View the existing /etc/environment file before editing.

Screenshot name: task4_etc_environment_before.png

```
tooba65@ubuntu:~$ sudo cat /etc/environment
[sudo] password for tooba65:
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
tooba65@ubuntu:~$
```

Show current PATH (single print) — this is not a grep, but keep it as a dedicated screenshot:

```
tooba65@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
tooba65@ubuntu:~$
```

2. Add a new variable named Class="CC-<yourClassName>" in /etc/environment.

Screenshot name: task4_etc_environment_edit_vim.png

```
Ubuntu 64-bit x
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-BSE-5B"
```

3. Re-login or open a new terminal. Display the variable and PATH to confirm changes.

Screenshot name: task4_echo_class_and_path.png

```
Expanded Security Maintenance for Applications is not enabled.

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tooba65@ubuntu:~$ echo $Class
CC-BSE-5B
tooba65@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
tooba65@ubuntu:~$
```

4. Create a “welcome” script in your home directory

Screenshot name: task4_welcome_create_and_chmod.png

```
tooba65@ubuntu:~$ cat > ~/welcome <<'EOF'
>
> #!/bin/bash
>
> echo "Welcome to Cloud Computing $USER"
>
> EOF
tooba65@ubuntu:~$ chmod +x ~/welcome
tooba65@ubuntu:~$ ls -l ~/welcome
-rwxrwxr-x 1 tooba65 tooba65 55 Nov  7 13:06 /home/tooba65/welcome
```

5. Run the script directly.

Screenshot name: task4_welcome_run_dot.png

```
tooba65@ubuntu:~$ cd ~
tooba65@ubuntu:~$ ./welcome
Welcome to Cloud Computing tooba65
tooba65@ubuntu:~$
```

6. Add home directory to PATH in ~/.bashrc.

Screenshot name: task4_bashrc_path_added.png

```
export DB_PASSWORD=toobass123
PATH=$PATH:~
-- INSERT --
```

7. Apply and test PATH change

Screenshot name: task4_welcome_run_anywhere.png

```
tooba65@ubuntu:~$ source ~/.bashrc
tooba65@ubuntu:~$ cd ~
tooba65@ubuntu:~$ welcome
Welcome to Cloud Computing tooba65
tooba65@ubuntu:~$
```

Task 5 – Block and Allow SSH Using UFW

Steps:

1. Enable UFW and show its status.

Screenshot name: task5_ufw_enable_and_status.png

```
tooba65@ubuntu:~$ sudo ufw enable
[sudo] password for tooba65:
Firewall is active and enabled on system startup
tooba65@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip
tooba65@ubuntu:~$
```

2. Block SSH port (22) and verify in status.

Screenshot name: task5_ufw_deny_22_and_status.png

```
tooba65@ubuntu:~$ sudo ufw deny 22/tcp
Rule updated
Rule updated (v6)
tooba65@ubuntu:~$ sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 22/tcp DENY IN Anywhere
[ 2] 22/tcp (v6) DENY IN Anywhere (v6)

tooba65@ubuntu:~$
```

3. Try SSH login from Windows — it should fail.

Screenshot name: task5_ssh_attempt_blocked.png

```
C:\ Command Prompt
Microsoft Windows [Version 10.0.19045.6159]
(c) Microsoft Corporation. All rights reserved.

C:\Users\malik>ssh tooba65@192.168.202.128
ssh: connect to host 192.168.202.128 port 22: Connection timed out

C:\Users\malik>
```

4. Allow SSH again and reload UFW.

Screenshot name: task5_ufw_allow_reload_status.png

```
tooba65@ubuntu:~$ sudo ufw allow 22/tcp
Rule updated
Rule updated (v6)
tooba65@ubuntu:~$ sudo ufw reload
Firewall reloaded
tooba65@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

5. Try SSH login again from Windows — it should succeed.

Screenshot name: task5_ssh_success_after_allow.png

```
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Fri Nov 7 01:23:21 PM UTC 2025

System load: 0.0 Processes: 234
Usage of /: 93.6% of 9.75GB Users logged in: 1
Memory usage: 10% IPv4 address for ens33: 192.168.202.128
Swap usage: 0%

=> / is using 93.6% of 9.75GB

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  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

$ updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Fri Nov 7 06:20:31 2025 from 192.168.202.1
tooba65@ubuntu:~$
```

Task 6 – SSH Key-Based Authentication

A. On Windows (Client Side — Git Bash)

Steps:

1. Generate a new SSH key pair on Windows (client machine).

Screenshot name: task6_windows_sshkey_and_list.png

```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ ssh-keygen -t ed25519 -f ~/.ssh/id_lab7 -C "lab_key"
Generating public/private ed25519 key pair.
Enter passphrase for "/c/Users/malik/.ssh/id_lab7" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/malik/.ssh/id_lab7
Your public key has been saved in /c/Users/malik/.ssh/id_lab7.pub
The key fingerprint is:
SHA256:akz3/Mz3DZXyd2ADCEiQ+RPz080CegEHCrkyLx/MkTA lab_key
The key's randomart image is:
+--[ED25519 256]--+
|.. o*...      |
|E. .+ = . .   |
|+. . + + . .   |
|+ o  = . . .   |
|. = . ..+S+ .+. |
|. = .o.o+oo .o+ |
| o . .+ oo .o|
| . .      + ..+|
|              +. .o|
+-----[SHA256]-----+
```

```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ ls -la ~/.ssh
total 33
drwxr-xr-x 1 malik 197609  0 Nov  7 09:21 ./
drwxr-xr-x 1 malik 197609  0 Oct 19 20:12 ../
-rw-r--r-- 1 malik 197609 116 Oct  3 12:28 config
-rw-r--r-- 1 malik 197609 419 Oct  3 08:22 id_ed25519
-rw-r--r-- 1 malik 197609 110 Oct  3 08:22 id_ed25519.pub
-rw-r--r-- 1 malik 197609 399 Nov  7 09:21 id_lab7
-rw-r--r-- 1 malik 197609  89 Nov  7 09:21 id_lab7.pub
-rw-r--r-- 1 malik 197609 1671 Oct  3 09:41 known_hosts
-rw-r--r-- 1 malik 197609 938 Oct  3 09:41 known_hosts.old
```

2. View the contents of the public key file.

Screenshot name: task6_windows_public_key.png

```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ cat ~/.ssh/id_lab7.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL9xqzrjrH1k6+acBgtKjEk6FTFuRhYGd0a4frjPRI5+
lab_key
```

3. On the server, check if .ssh folder and authorized_keys file exist.

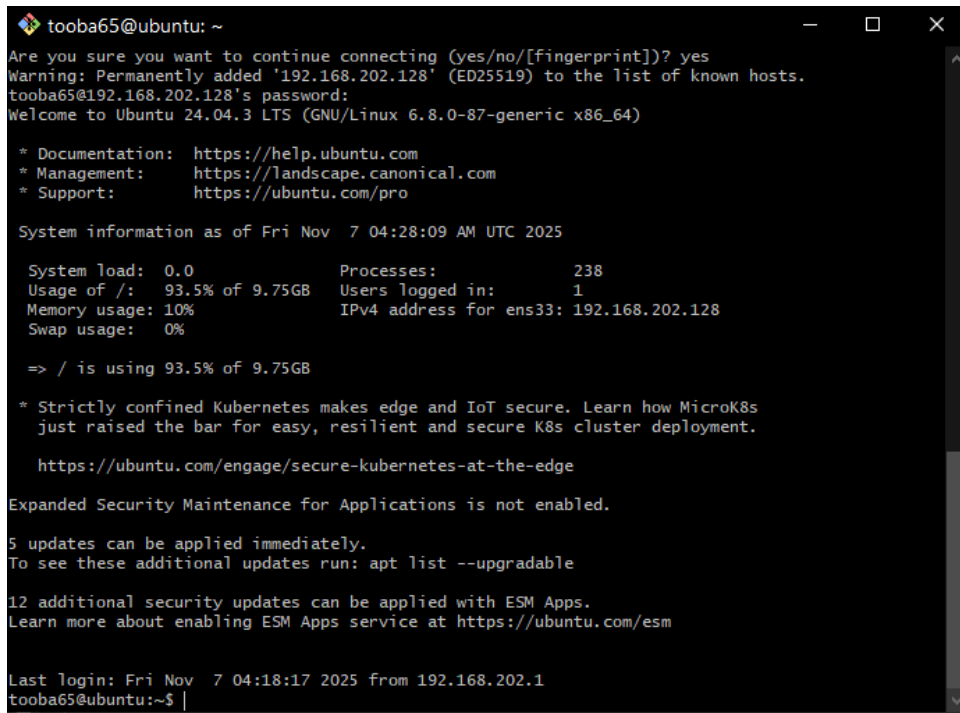
Screenshot name: task6_server_ssh_folder_and_authorized_before.png

```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ > ~/.ssh/known_hosts

malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ cat ~/.ssh/known_hosts
```

4: Connect normally with password to add host

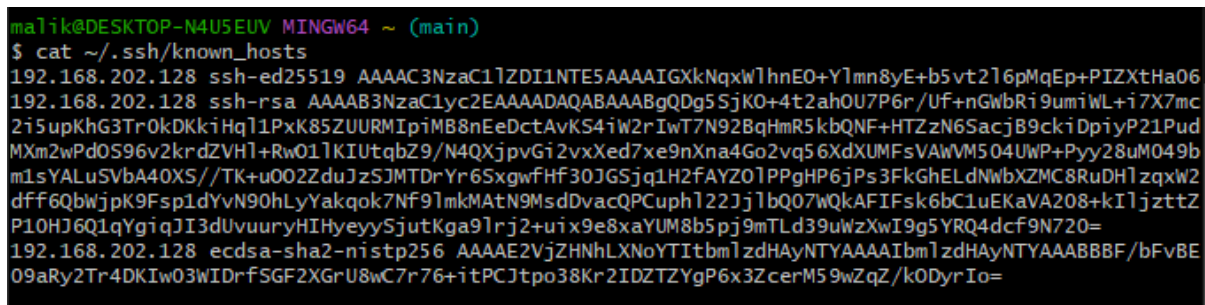
Screenshot: task6_windows_ssh_accept_hostkey_and_login.png



```
tooba65@ubuntu: ~  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.202.128' (ED25519) to the list of known hosts.  
tooba65@192.168.202.128's password:  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Fri Nov  7 04:28:09 AM UTC 2025  
  
System load:  0.0          Processes:            238  
Usage of /:   93.5% of 9.75GB Users logged in:      1  
Memory usage: 10%         IPv4 address for ens33: 192.168.202.128  
Swap usage:   0%  
  
=> / is using 93.5% of 9.75GB  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
5 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
12 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Fri Nov  7 04:18:17 2025 from 192.168.202.1  
tooba65@ubuntu:~$
```

5: Show known_hosts updated

Screenshot: task6_windows_known_hosts_after_connect.png

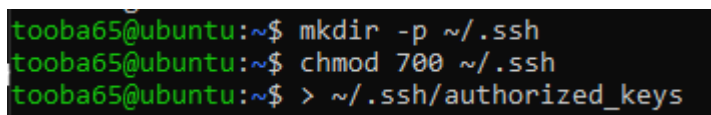


```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)  
$ cat ~/.ssh/known_hosts  
192.168.202.128 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGXkNqXw1hnEO+Y1mn8yE+b5vt216pMqEp+PIZXtHa06  
192.168.202.128 ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDg5SjKO+4t2ahOU7P6r/Uf+nGWbRi9umiWL+i7X7mc  
2i5upKhG3Tr0kDKkiHq11PxK85ZUURMipMB8nEeDctAvK54iW2rIwT7N92BqHmR5kbQNF+HTZnN6SaccjB9ckiDpiyP21Pud  
MXm2wPdOS96v2krdZVH1+RwO11KIUtqbZ9/N4QXjpvGi2vxXed7xe9nXna4Go2vq56XdXUMFsVAkVM504UWP+Pyy28uM049b  
m1sYALuSVbA40XS//TK+u002ZduJzSJMtdrYr6SxgwfHF30JG5jq1H2fAYZ01PPgHP6jPs3FkGhELdNwbXZMC8RuDH1zqxw2  
dff6QbwjP9Fsp1dYvN90hLyYakqok7Nf91mkMatN9MsDvacQPCuph122Jj1bQ07wQkAFIFsk6bC1uEKaVA208+kI1jztZ  
P10HJ6Q1qYgiqJI3dUvuuryHIHyeyySjutKga91rj2+uix9e8xaYUM8b5pj9mTLd39uWzXwI9g5YRQ4dcf9N720=  
192.168.202.128 ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBF/bFvBE  
09aRy2Tr4DKIw03WIDrfSGF2XGrU8wC7r76+itPCJtpo38Kr2IDZTZYgP6x3ZcerM59wZqZ/k0DyrIo=
```

B. On Ubuntu (Server Side — after logging in)

6: Prepare .ssh and clear old keys

Screenshot: task6_server_clear_authorized_keys.png



```
tooba65@ubuntu:~$ mkdir -p ~/.ssh  
tooba65@ubuntu:~$ chmod 700 ~/.ssh  
tooba65@ubuntu:~$ > ~/.ssh/authorized_keys
```

7: Add your Windows public key

Copy the **public key** line you saw in Step 2, then paste it into:

Screenshot: task6_server_add_key_and_show.png


```
tooba65@ubuntu:~$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL9xqzrjrH1k6+acBgtKjEk6FTFuRhYgD0a4frjPRI5+ lab_key" > ~/.ssh/authorized_keys
tooba65@ubuntu:~$ chmod 600 ~/.ssh/authorized_keys
tooba65@ubuntu:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL9xqzrjrH1k6+acBgtKjEk6FTFuRhYgD0a4frjPRI5+ lab_key
```

8. From Windows host test passwordless login (capture successful login in one screenshot):

Screenshot: task6_ssh_passwordless_login.png

```
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ ssh tooba65@192.168.202.128
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Nov  7 06:10:49 AM UTC 2025

System load:  0.0           Processes:            236
Usage of /:   93.5% of 9.75GB Users logged in:       1
Memory usage: 11%          IPv4 address for ens33: 192.168.202.128
Swap usage:   0%

=> / is using 93.5% of 9.75GB

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   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Fri Nov  7 05:56:15 2025 from 192.168.202.1
tooba65@ubuntu:~$
```

9. Also demonstrate explicit identity usage (single screenshot):

Screenshot: task6_ssh_with_identity_file.png

```
tooba65@ubuntu: ~
malik@DESKTOP-N4U5EUV MINGW64 ~ (main)
$ ssh -i ~/.ssh/id_lab7 tooba65@192.168.202.128
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Nov  7 05:53:33 AM UTC 2025

System load:  0.0           Processes:            236
Usage of /:   93.5% of 9.75GB Users logged in:       1
Memory usage: 11%          IPv4 address for ens33: 192.168.202.128
Swap usage:   0%

=> / is using 93.5% of 9.75GB

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Last login: Fri Nov  7 05:53:34 2025 from 192.168.202.1
tooba65@ubuntu:~$
```

Exam Evaluation Questions

Q1: Quick Environment Audit

- Objective: Demonstrate you can inspect the current environment and extract a few key variables.
- Actions & evidence:
 - i. Run a single command to display environment variables and capture its output.
 - Save screenshot: EE_q1_env_all.png

```
tooba65@ubuntu:~$ printenv
SHELL=/bin/bash
DB_PASSWORD=labpass123
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XDG_SEAT=seat0
PWD=/home/tooba65
LOGNAME=tooba65
XDG_SESSION_TYPE=tty
SYSTEMD_EXEC_PID=1221
DB_USER=labuser
HOME=/home/tooba65
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:
31:*.r2=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg
35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.og
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.
g-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memo
INVOCATION_ID=f3cd097b6abc4f00869b999238915a69
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=tooba65
SHLVL=1
XDG_VTNR=1
DB_URL=postgres://db.example.local:5432/mydb
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
```

- ii. In the same terminal session, run three filters (one per line) to show values for PATH, LANG, and PWD, then capture a single screenshot showing the three outputs together.
 - Save screenshot: EE_q1_env_filters.png

```
tooba65@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/tooba65/Desktop:/home/tooba65
tooba65@ubuntu:~$ echo $LANG
en_US.UTF-8
tooba65@ubuntu:~$ echo $PWD
/home/tooba65
```

Q2: Short-lived Student Info

- Objective: Show how temporary environment variables behave (session-scoped).
- Actions & evidence:
 - i. In one terminal, set three variables (STUDENT_NAME, STUDENT_ROLL_NUMBER, STUDENT_SEMESTER) using export — execute all three consecutively and capture them in one screenshot (show the commands executed).
 - Save screenshot: EE_q2_exports.png

```
tooba65@ubuntu:~$ export STUDENT_NAME="Tooba Shafique"
tooba65@ubuntu:~$ export STUDENT_ROLL_NUMBER="23F-1234"
tooba65@ubuntu:~$
tooba65@ubuntu:~$ export STUDENT_SEMESTER="5th"
```

- ii. Still in the same session, print the three values with echo (grouped) and capture the outputs in one screenshot.

- Save screenshot: EE_q2_echoes.png

```
tooba65@ubuntu:~$ echo $STUDENT_NAME
Tooba Shafique
tooba65@ubuntu:~$ echo $STUDENT_ROLL_NUMBER
23F-1234
tooba65@ubuntu:~$ echo $STUDENT_SEMESTER
5th
```

- iii. Use a single printenv|grep command to list any STUDENT_ variables and capture the result.

- Save screenshot: EE_q2_printenv_grep.png

```
tooba65@ubuntu:~$ printenv | grep STUDENT_
STUDENT_NAME=Tooba Shafique
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=23F-1234
tooba65@ubuntu:~$
```

- iv. Exit that shell, open a fresh terminal, and show that the STUDENT_ variables are not set (use echo and printenv|grep together) — capture in one screenshot.

- Save screenshot: EE_q2_after_restart.png

```
tooba65@ubuntu:~$ echo $STUDENT_NAME
tooba65@ubuntu:~$ printenv | grep STUDENT_
tooba65@ubuntu:~$
```

Q3: Make It Sticky (Persistence Check for Student Info)

- Objective: Demonstrate persistence of environment variables across sessions via shell configuration.
- Actions & evidence:
 - i. Edit ~/.bashrc and append the three STUDENT_* exports. Capture a screenshot of the editor showing the new lines.
 - Save screenshot: EE_q3_bashrc_editor.png

```
export STUDENT_NAME="Tooba Shafique"
export STUDENT_ROLL_NUMBER="23F-1234"
export STUDENT_SEMESTER="5th"
```

- ii. Reload your shell config with a single command and then verify the three variables and show `printenv | grep '^STUDENT_'` — capture these verification outputs together in one screenshot.

- Save screenshot: EE_q3_after_source.png

```
tooba65@ubuntu:~$ source ~/.bashrc
tooba65@ubuntu:~$ echo $STUDENT_NAME
Tooba Shafique
tooba65@ubuntu:~$
tooba65@ubuntu:~$ echo $STUDENT_ROLL_NUMBER
23F-1234
tooba65@ubuntu:~$
tooba65@ubuntu:~$ echo $STUDENT_SEMESTER
5th
```

```
tooba65@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Tooba Shafique
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=23F-1234
```

- iii. Close and re-open a terminal and demonstrate the `STUDENT_NAME` variable is available (echo and `printenv grep` together) — capture in one screenshot.

- Save screenshot: EE_q3_after_restart.png

```
tooba65@ubuntu:~$ echo $STUDENT_NAME
Tooba Shafique
tooba65@ubuntu:~$ printenv | grep STUDENT_
STUDENT_NAME=Tooba Shafique
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=23F-1234
tooba65@ubuntu:~$
```

Q4: Firewall Rules: Block and Restore Ping (ICMP)

- Objective: Demonstrate you can block ping (ICMP echo) traffic using `ufw` and then re-allow it; show effect from a client.
- Actions & evidence:
 - i. Enable `ufw` and capture the enable command and status together in one screenshot.
 - Save screenshot: EE_q5_ufw_enable_status.png

```
tooba65@ubuntu:~$ sudo ufw enable
[sudo] password for tooba65:
Firewall is active and enabled on system startup
tooba65@ubuntu:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
22/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)

- ii. Add a rule to block ping (ICMP echo) and show ufw status numbered in the same screenshot.

- Suggested command example:
- `sudo ufw deny proto icmp from any to any` `sudo ufw status numbered`
- Save screenshot: EE_q5_ufw_deny_ping_status.png

```
tooba65@ubuntu:~$ sudo ufw deny proto icmp from any to any
ERROR: Unsupported protocol 'icmp'
tooba65@ubuntu:~$ sudo ufw status numbered
Status: active
```

To	Action	From
--	-----	----
[1] 22/tcp	ALLOW IN	Anywhere
[2] 22/tcp (v6)	ALLOW IN	Anywhere (v6)

- iii. From your Windows host (or another client), attempt to ping the server while the rule is active and capture the blocked/failing ping in one screenshot.

- Save screenshot: EE_q5_ping_blocked.png

```
tooba65@ubuntu:~$ sudo ufw deny proto icmp from any to any
ERROR: Unsupported protocol 'icmp'
tooba65@ubuntu:~$ sudo ufw status numbered
Status: active
```

To	Action	From
--	-----	----
[1] 22/tcp	ALLOW IN	Anywhere
[2] 22/tcp (v6)	ALLOW IN	Anywhere (v6)

```
tooba65@ubuntu:~$ sudo sysctl -w net.ipv4.icmp_echo_ignore_all=1
net.ipv4.icmp_echo_ignore_all = 1
```

```
C:\Users\malik>ping 192.168.202.128

Pinging 192.168.202.128 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.202.128:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- iv. Re-allow ping (ICMP) (or remove the deny rule) and capture the allow/reload/status sequence in one screenshot.

- Suggested command example:
- `sudo ufw allow proto icmp from any to any`
- `sudo ufw reload sudo ufw status`
- Save screenshot: EE_q5_ufw_allow_ping_status.png

```
tooba65@ubuntu:~$ sudo sysctl -w net.ipv4.icmp_echo_ignore_all=0
net.ipv4.icmp_echo_ignore_all = 0
```

- v. From the client, ping the server again and capture successful replies in one screenshot.

- Save screenshot: EE_q5_ping_success.png

```
C:\Users\malik>ping 192.168.202.128

Pinging 192.168.202.128 with 32 bytes of data:
Reply from 192.168.202.128: bytes=32 time<1ms TTL=64
Reply from 192.168.202.128: bytes=32 time<1ms TTL=64
Reply from 192.168.202.128: bytes=32 time<1ms TTL=64
Reply from 192.168.202.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.202.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```